

A Digital-Twin Prototype for Personalised Multimodal Driver-Safety Risk Prediction: Evidence, Architecture and Evaluation Protocol

Ananya Shirshi, Swara Anakaram, Laxmi Shankari, Aashna Shaikh

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Project guidance: Mr. Akash Chatake, M.Tech — Artificial Intelligence and Machine Learning, BITS Pilani · R&D, Chatake Innworks Private Limited

Abstract. Driver-monitoring systems can misinterpret physiology when they apply generic thresholds without accounting for individual baseline, recent activity, environment, signal quality and driving context. This paper reports the repository evidence and research protocol for a smart-steering-wheel driver-safety prototype. The implemented artefacts include synthetic physiology generators, CSV telemetry replay, FastAPI services, a React dashboard, multiple legacy risk-inference paths, a versioned initial telemetry contract, and a browser-based digital twin. We propose a personalised, multimodal extension in which vision, physiology, context/habit and driving-behaviour features are temporally fused and evaluated against expected individual recovery trajectories. A reproducible repository-derived dataset study confirms that the primary generator file contains 90,000 balanced synthetic scenario samples and the replay file contains 296 rolling-window records. The paper deliberately distinguishes these software/data facts from unimplemented vision/context/temporal modules and from unperformed real-driver validation. The contribution is therefore an evidence-grounded architecture and validation protocol for non-diagnostic driver-safety decision support, not a claim of medical, fatigue or crash-prediction performance.

Keywords: driver monitoring; smart steering wheel; digital twin; multimodal fusion; uncertainty; physiological signal quality; safety decision support.

Highlights

- Reports an implemented FastAPI–React–digital-twin smart-wheel prototype and a versioned starting telemetry contract.
- Audits conflicting KNN, Random Forest and calibrated-XGBoost evidence rather than merging unsupported results.
- Proposes personalised recovery residuals, missing-modality handling and multi-horizon risk evaluation.
- Uses driver-wise, time-aware evaluation and calibration as publication gates.

1. Introduction

Smart steering wheels offer a practical location for measuring wheel interaction and selected physiological signals. Yet high heart rate, electrodermal activity (EDA), altered grip, and temperature change are ambiguous without context. They can reflect activity recovery, traffic stress, environmental conditions, sensing artefact, or a driver-safety concern. This work is motivated by the supplied smart-wheel baseline study [1], which is treated strictly as prior work. Unlike a fixed-range detector, the proposed approach estimates whether observations deviate from a driver-specific and context-conditioned expectation.

The project is limited to non-diagnostic decision support. It does not diagnose arrhythmia or disease, and it does not command an on-road vehicle. These limits are necessary because the present repository contains synthetic labels and prototype simulations, not clinical or real-road validation.

2. Materials and implemented system

The repository contains (i) FastAPI backend services and replay code, (ii) a React/Vite dashboard, (iii) synthetic data generators, (iv) model/inference code, and (v) a digital-twin MVP with WebSocket telemetry and a Three.js scene. The canonical `driver-twin.v1` JSON schema currently requires a schema version, event time, sequence, wheel angle, heart rate and risk state.

Artefact	Verified role	Evidence boundary
`backend/app/services/simulator.py`	Sequential replay of rolling HR, HRV proxy, GSR and temperature values.	Grip is fixed in this path; this is replay rather than live acquisition.
`AI_Module/AIML/dataset_generator/`	Generates 9 physiological/ECG features for 10 synthetic driver IDs.	Scenario generator, not participant data.
`digital-twin/`	Live local FastAPI/WebSocket visual MVP and scenario controls.	Explicitly not a vehicle-physics or medical model.

3. Dataset study and feature engineering

The primary analysed source is `AI\_Module/AIML/dataset\_generator/generated\_dataset.csv`: 90,000 rows $\times$ 12 columns, 10 synthetic driver IDs, and three equally represented scenario labels (30,000 rows per label). Hence the class proportion is 30,000 / 90,000 = 0.3333, or 33.33% for each class. The selected replay file has 296 rows $\times$ 6 columns and uses a five-sample rolling feature window.

$$\mu_t = (1/W) \sum_{i=t-W+1}^t x_i, \sigma_t = \sqrt{[(1/W) \sum_{i=t-W+1}^t (x_i - \mu_t)^2]}, W = 5.$$

The current preprocessing uses these statistics to create HR rolling mean, HR rolling standard deviation, GSR rolling mean and temperature rolling mean. The root-level 270,000-row CSV was excluded because its values are implausible for the declared units and require a data-provenance investigation.

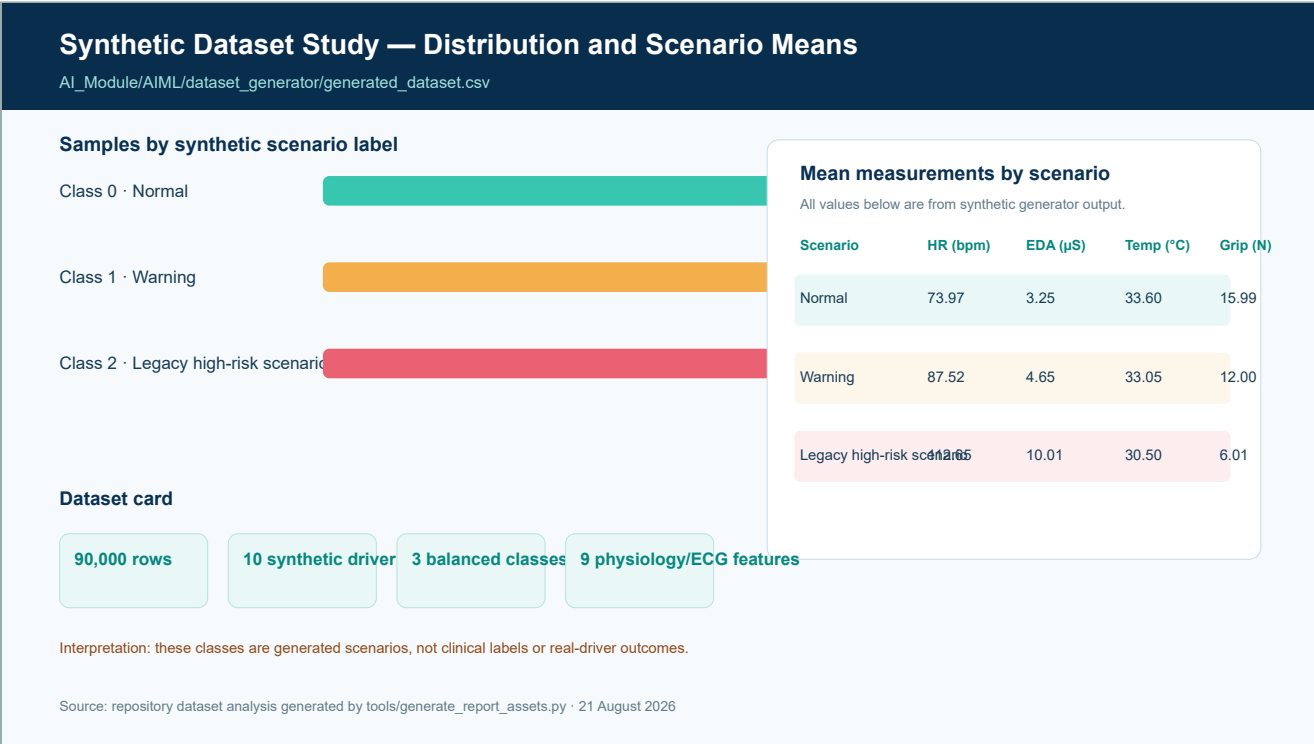


Fig. 1. Repository-derived synthetic generator dataset distribution and class-wise mean signals. The source generator uses a medical-sounding legacy label for class 2; it is treated here only as a synthetic high-risk scenario label.

Replay Dataset Study — Rolling Signal Trends

backend/data/processed_driver_features.csv

Rolling telemetry replay — 296 time-indexed rows

Heart rate (bpm)

min 73.69 · mean 85.11 · max 128.17



GSR / EDA (μS)

min 3.09 · mean 5.11 · max 13.83



Skin temperature (°C)

min 28.63 · mean 32.67 · max 33.65



Replay label composition

Class 0: 176 rows · Class 1: 40 rows · Class 2: 80 rows

The data was created from the project's synthetic scenario generator; trend shape is evidence of replay behaviour, not human physiology.

Source: repository dataset analysis generated by tools/generate_report_assets.py · 21 August 2026

Fig. 2. Repository-derived rolling telemetry replay trends (296 records). This validates the replay data path, not human physiology.

4. Risk inference evidence and proposed model

Three model lines coexist. A legacy KNN path consumes nine fields and stabilises predictions over three samples. A legacy report documents a Random Forest using four rolling features and 60 test samples, but its run/artifact cannot be reconciled with the current training source. The current training script creates a calibrated XGBoost binary classifier from nine generated features; the predictor mixes model probability with hand-written penalties and a five-second persistence average. Consequently, no historical accuracy value is presented as a paper result.

Model line	Feature set	Status
KNN	9 live physiology/ECG fields	Legacy, feature compatibility not proven.
Random Forest	4 rolling replay features	Historical report only; no reproducible matched run.
Calibrated XGBoost	9 generated physiology features	Current artifact path; synthetic-data evidence only.

The proposed temporal model is:

$$p_{t,h} = F_{\theta}(x_{1:t} - \hat{x}_{1:t} \ v_{1:t} \ b_{1:t} \ c_{1:t} \ q_{1:t} \ m_{1:t}), \ h \in \{0, 5, 10, 15 \text{ min}\}.$$

Here x is physiology/wheel telemetry, $\hat{x}$ is an expected personalised trajectory, v is vision, b is driving behaviour, c is context/habit, q is signal quality and m is a missing-modality mask. A TCN, LSTM, or compact Transformer is a future empirical choice, not an implemented model.

5. Digital-twin workflow

The MVP is a software integration platform. Its future technology flow is sensor/replay → validation (units, range, time, quality) → time-aligned feature fusion → versioned inference → persistence-aware alert policy → dashboard/twin and evidence store. A ROS 2/CARLA pathway is documented in the repository for deterministic scenario replay and hardware-in-the-loop testing [2].

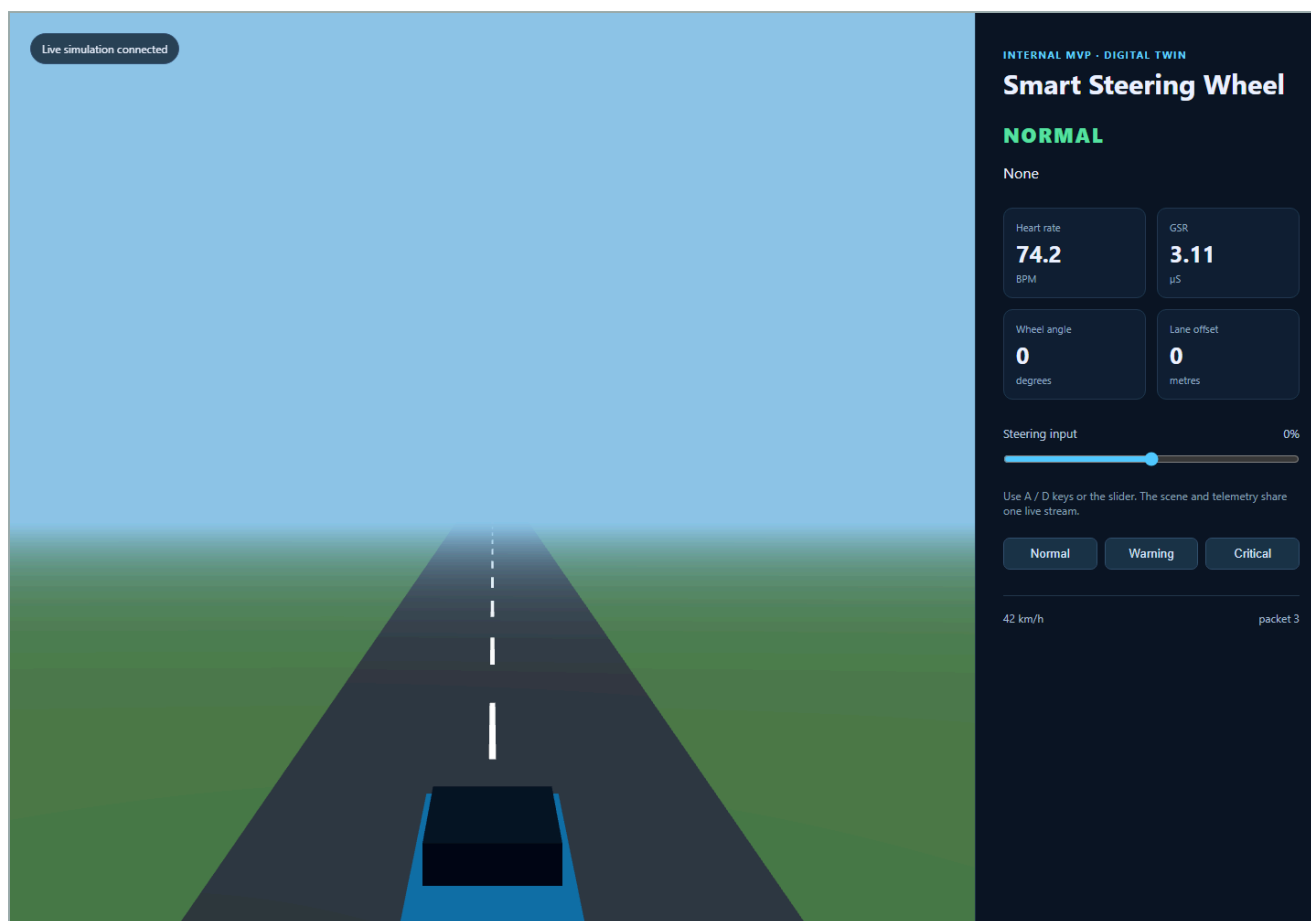


Fig. 3. Local execution screenshot of the actual browser digital-twin MVP. The dashboard cards and scene use an internal synthetic scenario stream.

6. Evaluation protocol

Evaluation must partition by driver and trip, not random frame/window rows. When personal adaptation is included, an early calibration segment is separated from scoring data. Baselines should compare vision-only, physiology/context-only, unconstrained fusion, full recovery-aware fusion, and missing-modality variants. Required metrics are macro F1, sensitivity, specificity, AUROC/AUPRC when suitable, Brier score, expected calibration error, false alerts per driving hour, alert lead time, latency and abstention coverage.

Publication gate. Real/simulator human data collection requires appropriate institutional ethics and consent processes; labels need a documented fatigue/impairment protocol. Synthetic generator labels alone cannot establish clinical, fatigue, or crash-risk performance.

7. Discussion, limitations and conclusion

The project demonstrates a credible integration foundation but not the proposed research method's performance. The absence of vision/context/behaviour datasets, temporal targets, driver-wise evaluation, calibration analysis and real-world validation prevents generalisable accuracy claims. The most valuable next step is to unify the feature/model contract and establish a governed dataset protocol before expanding the neural architecture.

In conclusion, the prototype provides a traceable route from smart-wheel telemetry to a driver-safety dashboard and digital twin. Its research novelty should be demonstrated through context-conditioned personal baselines, uncertainty-aware multimodal fusion and rigorous driver-wise testing—not through threshold labels or synthetic accuracy alone.

CRedit authorship contribution statement

Ananya Shirshi, Swara Anakaram, Laxmi Shankari and Aashna Shaikh: software investigation, documentation, dataset study, system analysis and report preparation. Mr. Akash Chatake: project guidance, research direction and technical supervision.

Declaration of competing interests

The authors declare no competing interests for this academic internship project.

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